

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

03

10 answers · Rationale-rich · Teach from doc

Q1**B****B.** 10

Choice B is correct. The number that's 8 more than x can be represented by the expression $x + 8$. It's given that the product of x and $x + 8$ is 180, so it follows that $xx + 8 = 180$, or $x^2 + 8x = 180$. Subtracting 180 from each side of this equation yields $x^2 + 8x - 180 = 0$. Factoring the left-hand side of this equation yields $x - 10x + 18 = 0$. Applying the zero product property to this equation yields two solutions: $x = 10$ and $x = -18$. Since x is a positive number, the value of x is 10.

Choice A is incorrect. If $x = 5$, the product of x and the number that's 8 more than x would be 513, or 65, not 180.

Choice C is incorrect. This is the value of the number that's 8 more than x , not the value of x .

Choice D is incorrect. If $x = 36$, the product of x and the number that's 8 more than x would be 3644, or 1,584, not 180.

Q2**D****D.** $w + 9$

Choice D is correct. It's given that the expression $ww + 9$ represents the area, in square centimeters, of a rectangular cutting board, where w is the width, in centimeters, of the cutting board. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the cutting board is represented by the expression $w + 9$.

Choice A is incorrect. This expression represents the area, in square centimeters, of the cutting board, not its length, in centimeters.

Choice B is incorrect. This expression represents the width, in centimeters, of the cutting board, not its length.

Choice C is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the cutting board, not its length, in centimeters.

Q3**C****C.** $x - 1$

Choice C is correct. It's given that f is a polynomial function and the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. A point in the form $(a, 0)$ on the graph of $f(x)$ means that $x - a$ is a factor of $f(x)$. Since the graph of the function f passes through the point $(1, 0)$, $x - 1$ must be a factor of $f(x)$.

Choice A is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-1, 0)$.

Choice B is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-4, 0)$.

Choice D is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(5, 0)$.

Q4**-6**

Accept also: 4

The correct answer is either -6 or 4. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. Thus, for an x -intercept of the graph of $y = f(x)$, $0 = f(x)$. Substituting 0 for $f(x)$ in the equation $f(x) = x + 6x - 4$ yields $0 = x + 6x - 4$. By the zero product property, $x + 6 = 0$ and $x - 4 = 0$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the x -coordinates of the x -intercepts of the graph of $y = f(x)$ are -6 and 4. Note that -6 and 4 are examples of ways to enter a correct answer.

Q5

C

C. $n = 6041 + 0.004^t$

Choice C is correct. It's given that the relationship between t and n is exponential. The table shows that the value of n increases as the value of t increases. Therefore, the relationship between t and n can be represented by an increasing exponential equation of the form $n = a1 + b^t$, where a and b are positive constants. The table shows that when $t = 0$, $n = 604$. Substituting 0 for t and 604 for n in the equation $n = a1 + b^t$ yields $604 = a1 + b^0$, which is equivalent to $604 = a1$, or $604 = a$. Substituting 604 for a in the equation $n = a1 + b^t$ yields $n = 6041 + b^t$. The table also shows that when $t = 1$, $n = 606.42$. Substituting 1 for t and 606.42 for n in the equation $n = 6041 + b^t$ yields $606.42 = 6041 + b^1$, or $606.42 = 6041 + b$. Dividing both sides of this equation by 604 yields approximately $1.004 = 1 + b$. Subtracting 1 from both sides of this equation yields that the value of b is approximately 0.004. Substituting 0.004 for b in the equation $n = 6041 + b^t$ yields $n = 6041 + 0.004^t$. Therefore, of the choices, choice C best represents the relationship between t and n .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

B

- B. The predicted body mass of the animal was approximately 362 kg 330 days after it was born.
-

Choice B is correct. In the statement " $m(330)$ is approximately equal to 362," the input of the function, 330, is the value of t , the elapsed time, in days, since the animal was born. The approximate value of the function, 362, is the predicted body mass, in kilograms, of the animal after that time has elapsed. Therefore, the predicted body mass of the animal was approximately 362 kg 330 days after it was born.

Choice A is incorrect. This would be the best interpretation of the statement " $m(362)$ is approximately equal to 330."

Choice C is incorrect. The number $\frac{330}{7}$ is the number of weeks, not the number of days, after the animal was born.

Choice D is incorrect. This would be the best interpretation of the statement " $m(362)$ is approximately equal to $\frac{330}{7}$."

Q7

A

- A. The initial height, in feet, of the object
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Choice A is correct. The variable t represents the seconds after the object is launched. Since $h(0) = 72$, this means that the height, in feet, at 0 seconds, or the initial height, is 72 feet.

Choices B, C, and D are incorrect and may be the result of misinterpreting the function in context.

Q8**C**

C. $p(t) = 24(1.162)^t$

Choice C is correct. This exponential growth model can be written in the form $p(t) = A(1+r)^t$, where $p(t)$ is the population t years after 1788, A is the initial population, and r is the yearly growth rate, expressed as a decimal. Since there were 24 jackrabbits in Australia in 1788, $A = 24$. Since the number of jackrabbits increased by an average of 16.2% each year, $r = 0.162$. Therefore, the equation that best models this situation is $p(t) = 24(1.162)^t$.

Choices A, B, and D are incorrect and may result from misinterpreting the form of an exponential growth model.

Q9**C**

C. 27

Choice C is correct. The function f is quadratic, so it will have either a maximum or a minimum at the vertex of the graph. Since the coefficient of the quadratic term (-0.28) is negative, the vertex will be at a maximum. The equation $f(\theta) = -0.28(\theta - 27)^2 + 880$ is given in vertex form, so the vertex is at $\theta = 27$. Therefore, an angle of 27 degrees gives the greatest airflow.

Choices A and B are incorrect and may be the result of misidentifying which value in a quadratic equation in vertex form represents the vertex. Choice D is incorrect. This choice identifies the maximum value of $f(\theta)$ rather than the value of θ for which $f(\theta)$ is maximized.

Q10

A

A. If the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft².

Choice A is correct. The function f gives the area of the rectangle, in ft², if its width is w ft. Since the value of $f(14)$ is the value of $f(w)$ if $w = 14$, it follows that $f(14) = 1,176$ means that $f(w)$ is 1,176 if $w = 14$. In the given context, this means that if the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft².

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from interpreting $f(w)$ as the width, in ft, of the rectangle if its area is w ft², rather than as the area, in ft², of the rectangle if its width is w ft.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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